Sentiment Analysis of Animated Series Using Natural Language Processing

Submitted in partial fulfillment of the requirements of the

Mini-Project of Natural Language Processing Lab for

Fourth Year of

Bachelors of Engineering

by

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CERTIFICATE

This is to certify that the mini-project entitled Sentiment Analysis of Animated Series Using Natural Language Processing is a bonafide work of Shaikh Hassaan Shakeel - 56 submitted to the University of Mumbai in partial fulfillment of the requirement for the Mini-Project of Natural Language Processing Lab for Fourth Year of the Bachelor of Engineering in “Artificial Intelligence & Data Science”.

Guide:

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Prof. Vijay Shankar

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Sentiment Analysis on Animated Series is a Natural Language Processing (NLP) mini-project that aims to classify viewer comments into three sentiment categories: Positive, Negative, and Neutral. The project focuses on comments related to two animated series, Hazbin Hotel and The Amazing Digital Circus. It addresses the challenge of automatically interpreting opinions expressed in informal text, where slang, ambiguity, and mixed emotions can make sentiment classification difficult.

The system uses text preprocessing techniques, including lowercasing, URL and mention removal, and whitespace normalization, to prepare the comments for analysis. The dataset is divided into training and testing sets using a stratified 80:20 split. Random oversampling is applied exclusively to the training data to balance the representation of the three sentiment classes. The processed text is converted into numerical features using Term Frequency–Inverse Document Frequency (TF-IDF), and a Logistic Regression classifier is trained to predict sentiment.

The expanded dataset contains 215 labelled comments, including 40 synthetically generated examples added to increase the representation of the Negative and Neutral classes. The final model, using unigram TF-IDF features and Logistic Regression with a regularization parameter of C = 0.5, achieved an accuracy of 53.49% on the held-out test set. Positive sentiment was classified more effectively than the other classes, while Neutral sentiment remained a significant challenge.

The trained model was integrated into a web-based application developed using Streamlit and deployed on Streamlit Community Cloud. The application allows users to enter viewer comments and obtain sentiment predictions through an interactive interface. The project demonstrates the practical application of NLP and machine learning in opinion classification while highlighting the importance of dataset size, annotation quality, and contextual understanding. Future improvements may include collecting a larger dataset of genuine viewer comments, refining sentiment annotations, and evaluating more advanced NLP models.

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Chapter 1

Aim

To develop a Natural Language Processing (NLP)-based sentiment analysis system that automatically classifies viewer comments related to animated series into three categories: Positive, Negative, and Neutral. The system aims to analyze opinions expressed in comments related to Hazbin Hotel and The Amazing Digital Circus using text preprocessing, TF-IDF feature extraction, and Logistic Regression.

The project also aims to develop an interactive web application using Streamlit, allowing users to enter viewer comments and obtain sentiment predictions. Through this implementation, the project demonstrates the practical application of NLP and machine learning techniques for automated opinion classification.

Chapter 2

Problem Statement

With the growing popularity of animated series and online discussion platforms, viewers frequently express their opinions through comments and reviews. These comments provide useful information about audience reactions to characters, animation, storylines, and episodes. However, manually analysing a large number of comments can be time-consuming and may make it difficult to identify overall audience sentiment efficiently.

Automatically classifying these comments presents several challenges. Viewer comments often contain informal language, slang, ambiguous expressions, and mixed opinions. A comment may express both positive and negative views, making it difficult to determine its overall sentiment. In particular, distinguishing neutral comments from mildly positive or negative comments can be challenging for a machine learning model.

The project aims to demonstrate the feasibility of an automated sentiment classification system while examining its performance and limitations on a relatively small dataset of labelled comments from Hazbin Hotel and The Amazing Digital Circus.

Chapter 3

Brief Theory

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to process, analyse, and interpret human language. It is used in various applications, including chatbots, machine translation, text classification, and sentiment analysis. NLP helps transform unstructured textual data into a form that computer systems can analyse and use to perform specific tasks.

Sentiment analysis is an NLP technique used to identify the opinion or emotional tone expressed in a piece of text. It is commonly applied to online reviews, social media comments, and customer feedback to determine whether an opinion is positive, negative, or neutral.

In this project, NLP is used to analyse viewer comments related to the animated series Hazbin Hotel and The Amazing Digital Circus. The system processes the text by converting it to lowercase, removing URLs and user mentions, and normalising whitespace. The cleaned comments are then converted into numerical features using Term Frequency–Inverse Document Frequency (TF-IDF), which represents words according to their importance within the dataset.

A Logistic Regression classifier is trained on these features to categorise comments into Positive, Negative, and Neutral sentiments. The trained model is integrated into a Streamlit web application, allowing users to enter comments and receive sentiment predictions through an interactive interface.

Thus, the project demonstrates how NLP and machine learning can be combined to automate the analysis of textual opinions. It also highlights the challenges of interpreting informal language and distinguishing neutral sentiment from comments expressing positive or negative opinions.

Chapter 4

Implementation Explanation

The Sentiment Analysis on Animated Series project is implemented as a web application that accepts viewer comments and classifies their overall sentiment as Positive, Negative, or Neutral. The system uses Natural Language Processing techniques and a machine learning model trained on labelled comments related to Hazbin Hotel and The Amazing Digital Circus.

The implementation consists of the following steps:

1. Data Collection: A dataset containing labelled viewer comments is prepared for training and evaluating the sentiment analysis model. The expanded dataset contains 215 comments across the three sentiment categories.

2. Text Preprocessing: The comments are cleaned by converting text to lowercase, removing URLs and user mentions, and normalising whitespace. This helps maintain consistency in the text supplied to the model.

3. Data Splitting and Balancing: The dataset is divided into training and testing sets using a stratified 80:20 split. Random oversampling is applied only to the training data to balance the three sentiment classes.

4. Feature Extraction: The cleaned comments are converted into numerical representations using Term Frequency–Inverse Document Frequency (TF-IDF). The implementation uses unigram features to represent individual words.

5. Model Training: A Logistic Regression classifier is trained using the TF-IDF features. The final configuration uses a regularisation parameter of C = 0.5 and a maximum of 1,000 iterations.

6. Model Evaluation: The trained model is evaluated using the held-out test set. Accuracy, precision, recall, F1-score, and a confusion matrix are used to examine its classification performance.

7. Web Application and Prediction: The trained sentiment analysis pipeline is integrated into a Streamlit web application and deployed using Streamlit Community Cloud. Users can enter a viewer comment and click the Analyze Sentiment button to receive the predicted sentiment through the interface.

The application is developed using Python and libraries including Pandas, Scikit-learn, and Streamlit. The implementation demonstrates how NLP and machine learning techniques can be combined to create an interactive sentiment classification system.

The application has been deployed to https://sentiment-animated.streamlit.app/

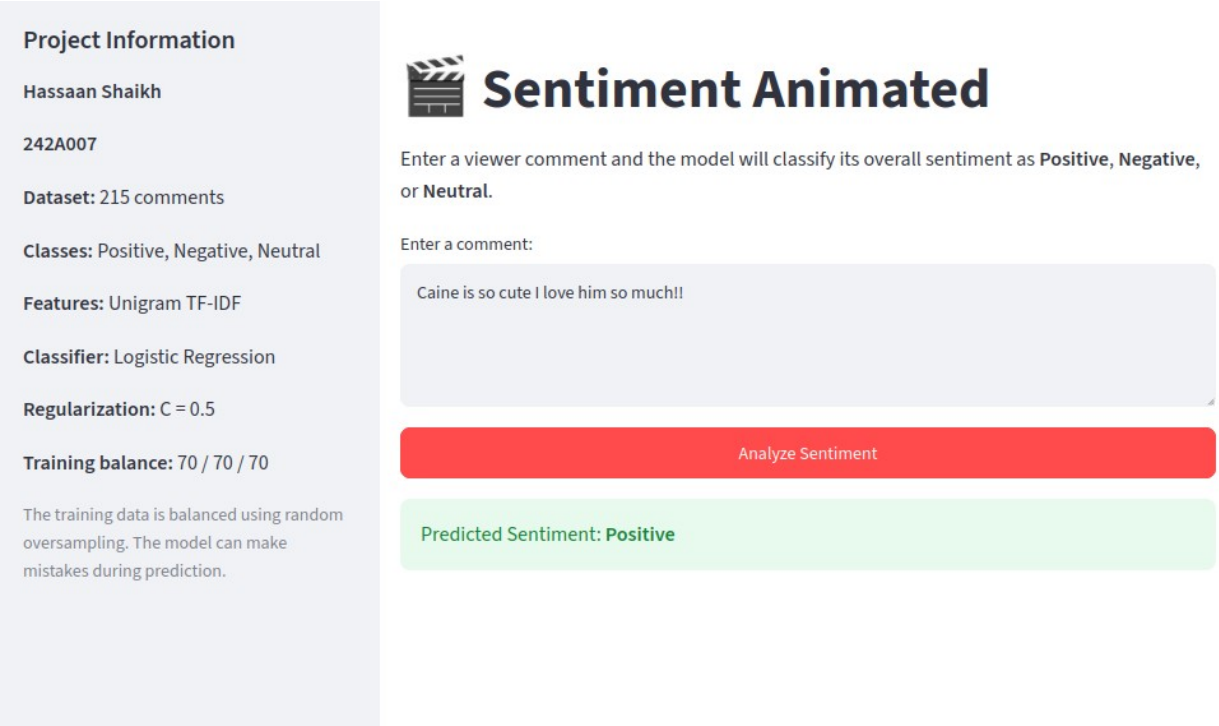


Fig. 1: The application interface and sentiment prediction

Chapter 5

Results

The Sentiment Analysis on Animated Series project was evaluated using a test dataset containing 43 comments. The model classifies viewer comments into three categories: Positive, Negative, and Neutral.

The final model uses unigram TF-IDF for feature extraction and Logistic Regression for classification, with the regularisation parameter set to C = 0.5.

The model achieved an overall accuracy of 53.49%. Its performance varied across the three sentiment categories:

- **Positive:** F1-score of 0.67
- **Negative:** F1-score of 0.50
- **Neutral:** F1-score of 0.33

The model performed best when identifying Positive comments, while Neutral comments were more difficult to classify correctly. This indicates that the model can identify some sentiment patterns but may struggle with ambiguous or informal viewer comments.

The project also includes a web application developed using Streamlit and deployed on Streamlit Community Cloud. Users can enter a comment and click the Analyze Sentiment button to obtain a predicted sentiment category.

The results demonstrate that TF-IDF and Logistic Regression can be used for basic sentiment classification. However, the accuracy is limited by the relatively small dataset, which contains 215 labelled comments, including 40 synthetically generated examples.

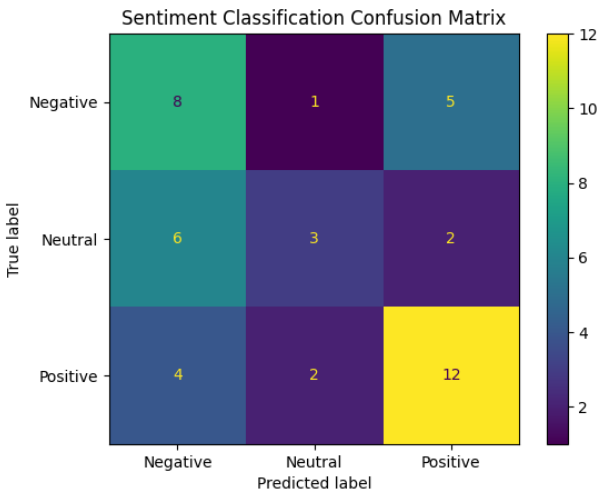


Fig. 2: Sentiment Confusion Matrix

Chapter 6
Conclusion

The project successfully demonstrates the use of Natural Language Processing and machine learning to classify viewer comments on Hazbin Hotel and The Amazing Digital Circus into Positive, Negative, and Neutral categories.

The system uses text preprocessing, TF-IDF feature extraction, and a Logistic Regression classifier. These components are integrated into a Streamlit web application, allowing users to enter comments and receive sentiment predictions through a simple interface.

The model achieved an accuracy of 53.49%, showing that it can identify sentiment patterns but still has room for improvement. Positive comments were classified more effectively than Neutral comments. The limited dataset size, synthetic examples, informal language, and ambiguity of viewer opinions are important limitations.

In the future, the system could be improved by collecting a larger dataset of genuine viewer comments, improving the quality and consistency of sentiment labels, and experimenting with more advanced NLP models. These improvements could help the application produce more reliable predictions across different types of comments.

Overall, the project provides a practical demonstration of how NLP techniques can be applied to sentiment analysis and deployed as an interactive web application.

Chapter 7

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